

from the same storage pool as the file system from which they were created.

- Recursive snapshots are created quickly as one atomic operation. The snapshots are created together (all at once) or not created at all. The benefit of atomic snapshot operations is that the snapshot data is always taken at one consistent time, even across descendent file systems.

ZFS Terminology	
alternate boot environment	A boot environment that is created by the <code>lucreate</code> command and possibly updated by the <code>luupgrade</code> command, but it is not currently the active or primary boot environment. The alternate boot environment (ABE) can be changed to the primary boot environment (PBE) by running the <code>luactivate</code> command.
checksum	A 256-bit hash of the data in a file system block. The checksum capability can range from the simple and fast <code>fletcher4</code> (the default) to cryptographically strong hashes such as <code>SHA256</code> .
clone	A file system whose initial contents are identical to the contents of a snapshot.
dataset	A generic name for the following ZFS entities: clones, file systems, snapshots, or volumes. Each dataset is identified by a unique name in the ZFS namespace. Datasets are identified using the following format: pool/path[@snapshot] pool Identifies the name of the storage pool that contains the dataset path Is a slash-delimited path name for the dataset object snapshot Is an optional component that identifies a snapshot of a dataset
file system	A ZFS dataset of type <code>filesystem</code> that is mounted within the standard system namespace and behaves like other file systems.
mirror	A virtual device that stores identical copies of data on two or more disks. If any disk in a mirror fails, any other disk in that mirror can provide the same data.
pool	A logical group of devices describing the layout and physical characteristics of the available storage. Space for datasets is allocated from a pool.
primary boot environment	A boot environment that is used by the <code>lucreate</code> command to build the alternate boot environment. By default, the primary boot environment (PBE) is the current boot environment. This default can be overridden by using the <code>lucreate -s</code> option.
RAID-Z	A virtual device that stores data and parity on multiple disks, similar to RAID-5 (RAID stands for Redundant Array of Independent Disks)
resilvering	The process of transferring data from one device to another device is known as <code>resilvering</code> . For example, if a mirror component is replaced or taken offline, the data from the up-to-date mirror component is copied to the newly restored mirror component. This process is referred to as mirror resynchronization in traditional volume management products.
virtual device	A logical device in a pool, which can be a physical device, a file, or a collection of devices.
volume	A dataset used to emulate a physical device (like a hard disk). For example, you can create a ZFS volume as a swap device.